Design Process Planning and Management

LAB EX-03

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EX03 - Thermal stress analysis of bimetallic strip

Target Date: 11.59 PM on 08 Feb

AIM

Perform thermal stress analysis of bimetallic strip made of steel and brass for switching actions and estimate the deflections for the given thermal environment

Tool : Fusion 360

Approach: Finite element simulation

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Simulation Input:

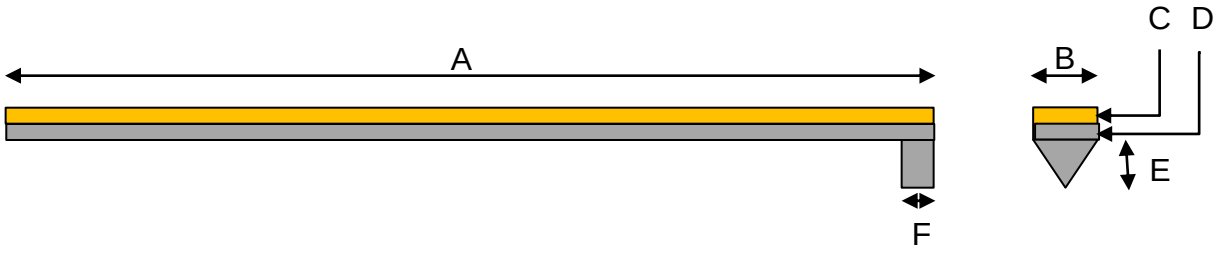
Rectangular bimetallic strip made up of steel(lower strip) and brass(top strip) per dimensions given fig.

Geometric dimensions are in mm as shown in fig

Boundary conditions:

Thermal environment
Body temperature of both strip 150 deg C

Structural:
Analysis A: Ux, Uy, Uz constrained at left faces of both strip
Analysis B: Ux, Uy, Uz constrained at both left and right faces of both strip
(**Remove triangular probe for this analysis**)



A= 100 mm
B=10 mm
C= 2 mm
D= 2 mm
E= 10 mm
F=3 mm

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Procedure

1. Create geometric model of two strips as two separate body or component
2. Save model EX03_ Roll No
3. Go to simulation menu
4. Make analysis setup – Thermal stress analysis
5. Assign material (Brass and steel) to two bodies
6. Define loads
   - Apply structural constraints: All 3 DOF of two left side faces are constrained
   - Apply body temperature of 150 deg C for both bodies

7. Contact generation between two bodies at the interfaces – Use option of automatic contact generation
8 Define meshing parameters with the average size as 4mm
9 Generate mesh
10 Check the model for input definition completeness
11 Solve the model
12 Post process the model to show deflection and stress pattern, Extract the max deflections at the probe tip
13 Save the model
14 Repeat the steps with 2mm and 1mm mesh size
15 Apply structural BC on right faces of bimetallic strip
16 Post process the results for stress and deflections
17 Compare the stresses and deflection for both load cases
18 Write the inferences

Upload the PDF of this completed report in Fusion 360 and VTOP.
All the models version to be stored in your folder

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Table 1: Thermal simulation results summary

Analysis case	BC	Elem. size (mm)	Max deflection at Probe Tip			Max/Min Stress on Brass strip (MPa)		Max/Min Stress on Steel strip (MPa)	
			Ux	Uy	Uz	Sxx	von-Mises	Sxx	von-Mises
1	Left side faces of strip constrained all DOF	4	0.2201 mm	0.003 mm	0.02164 mm	326.6 MPa	708.6 MPa	-275.7 MPa	4.634MPa
2	Left side faces of strip constrained all DOF	2	0.1959 mm	0.0027 mm	0.01551 mm	351.2 MPa	1040 MPa	-540.1 MPa	0.1058 MPa
2	Left side faces of strip constrained all DOF	1	0.2127 mm	0.002886 mm	0.07013 mm	410.5 MPa	1171 MPa	-1009 MPa	0.05065 MPa
4	Left and right side faces of strip constrained all DOF	1	0.01745 mm	0.007929 mm	0.00334 mm	226.4 MPa	84.93 MPa	-1275 MPa	1695 MPa

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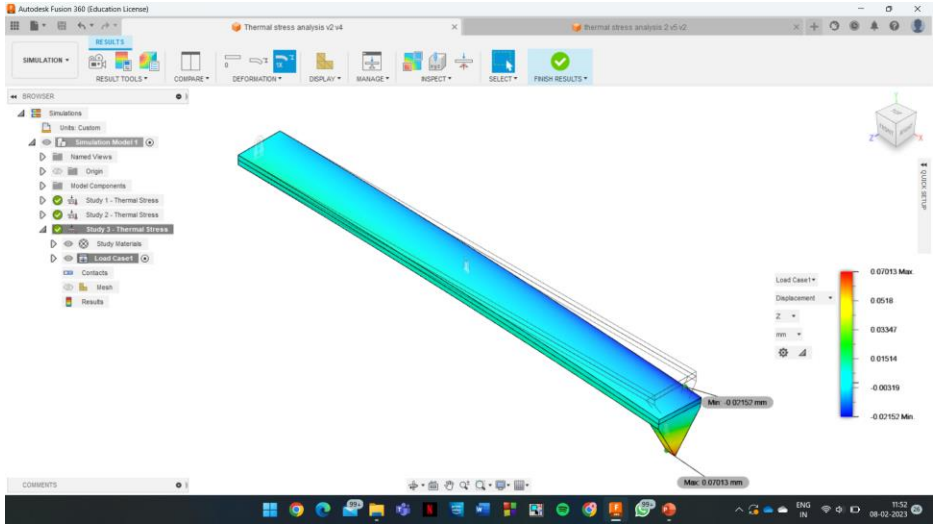
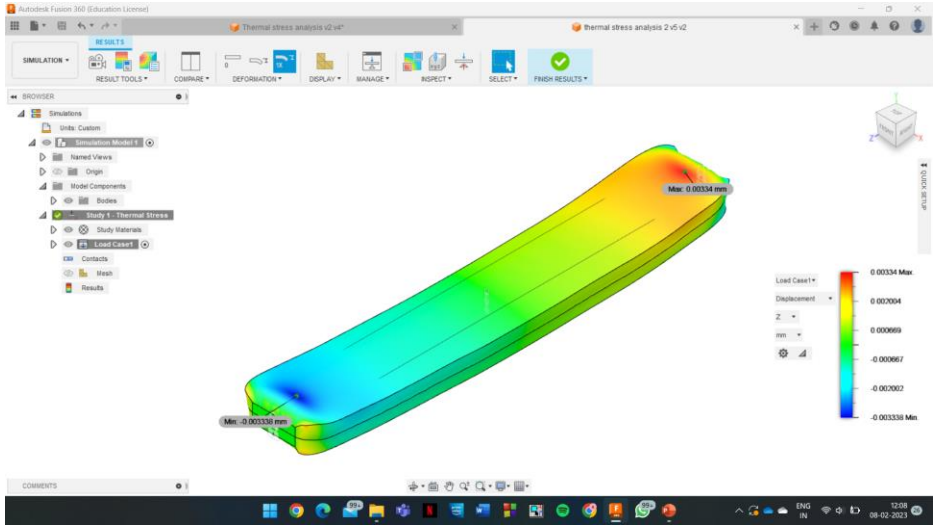
Deflection UZ distribution Pattern comparison between Load case 1 & 2



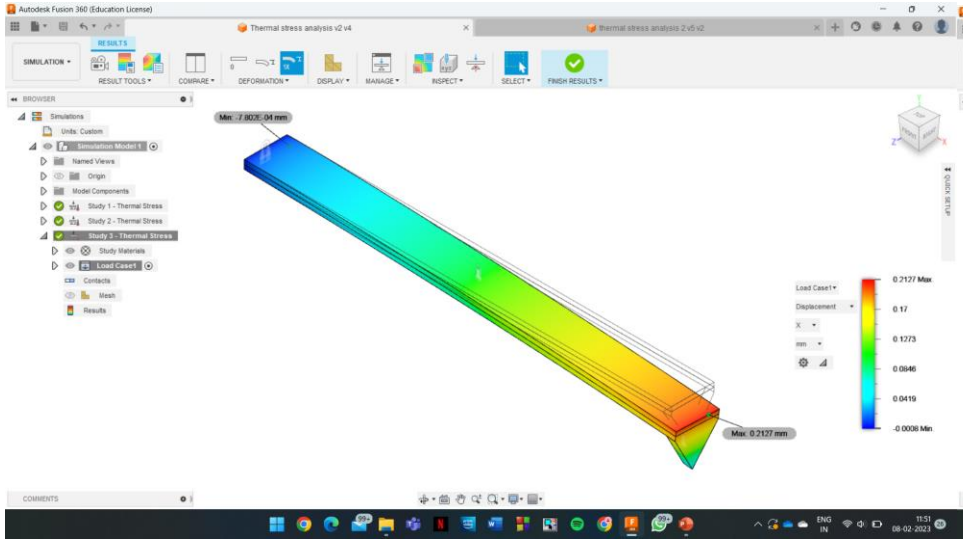
Load case 1



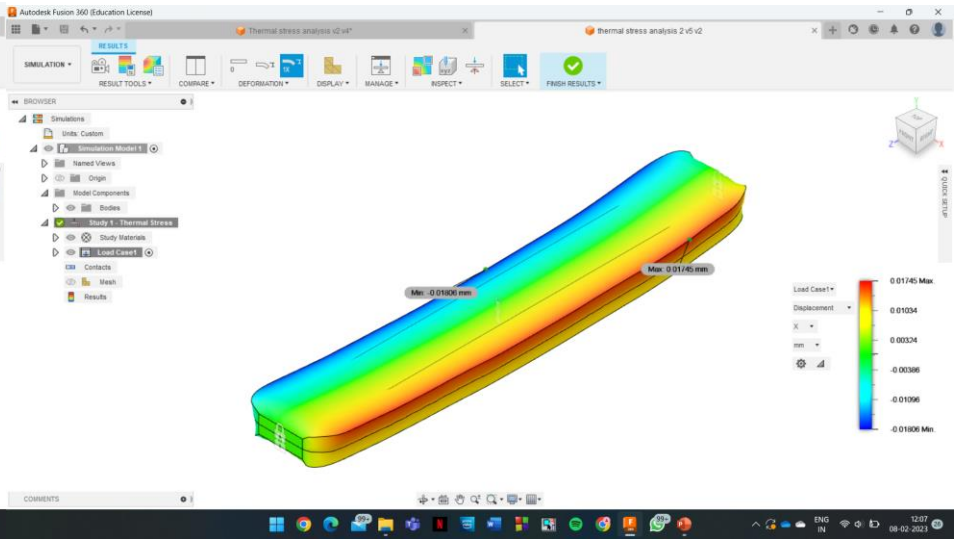
Load case 2

EX03 - Thermal stress analysis of bimetallic strip

Deflection Ux distribution Pattern comparison between Load case 1 & 2



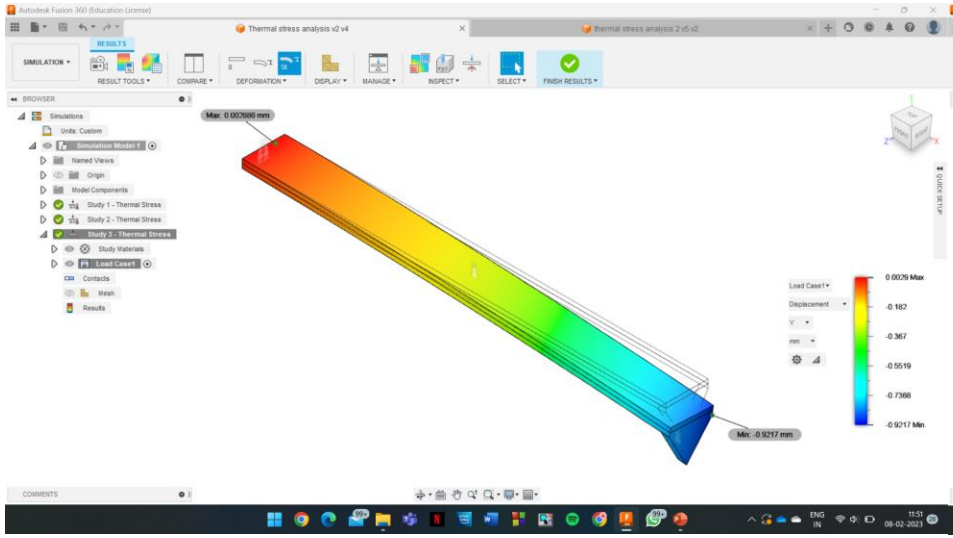
Load case 1



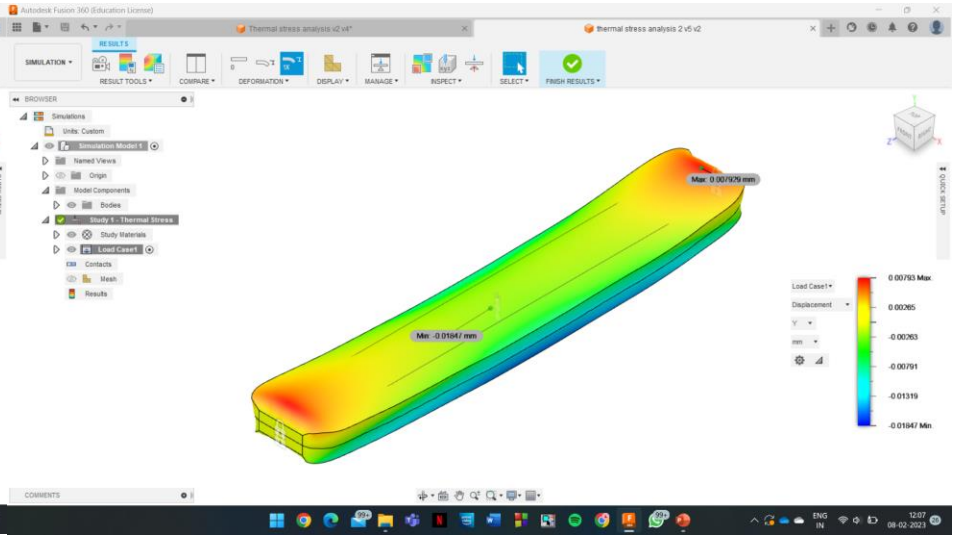
Load case 2

EX03 - Thermal stress analysis of bimetallic strip

Deflection Uy distribution Pattern comparison between Load case 1 & 2 (Indicating Max/Min value)



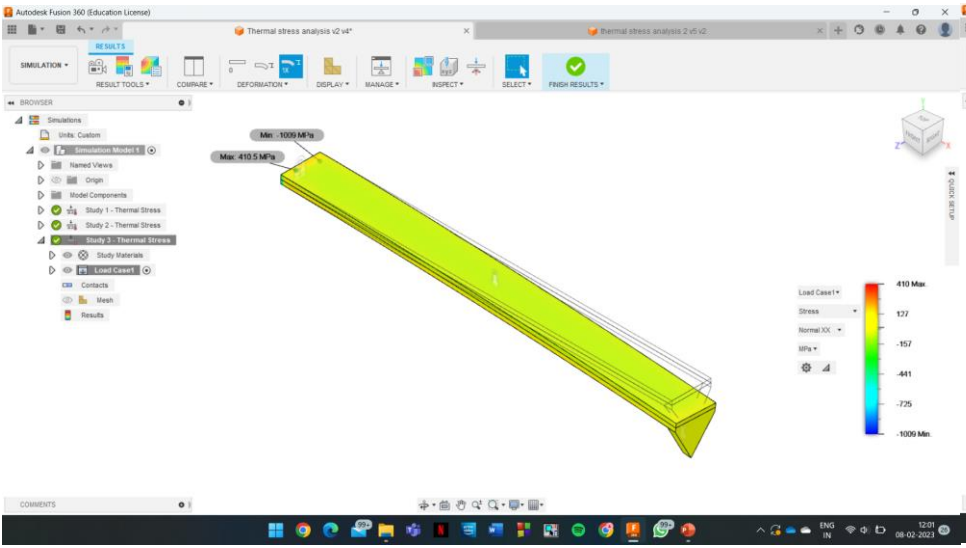
Load case 1



Load case 2

EX03 - Thermal stress analysis of bimetallic strip

Sxx distribution Pattern comparison between Load case 1 & 2 (Indicating Max/Min value)



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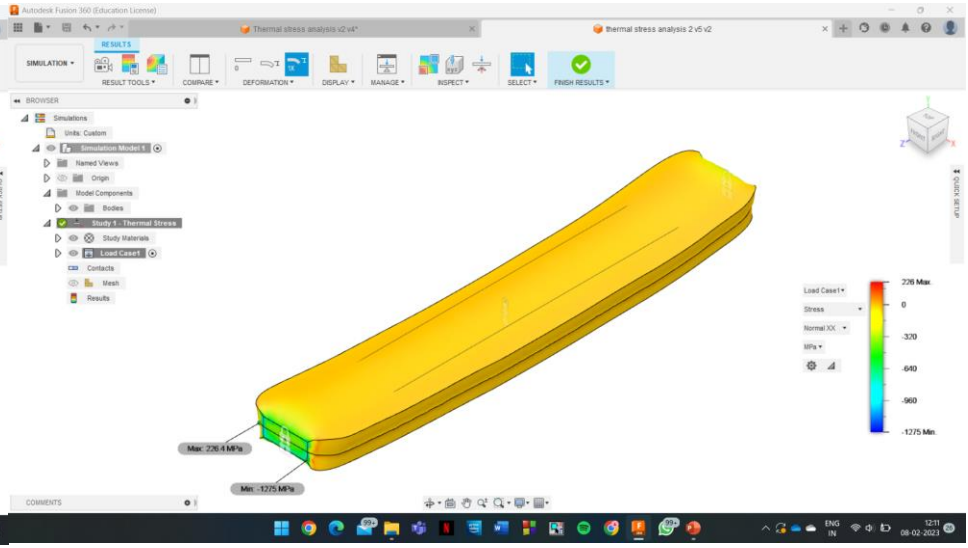
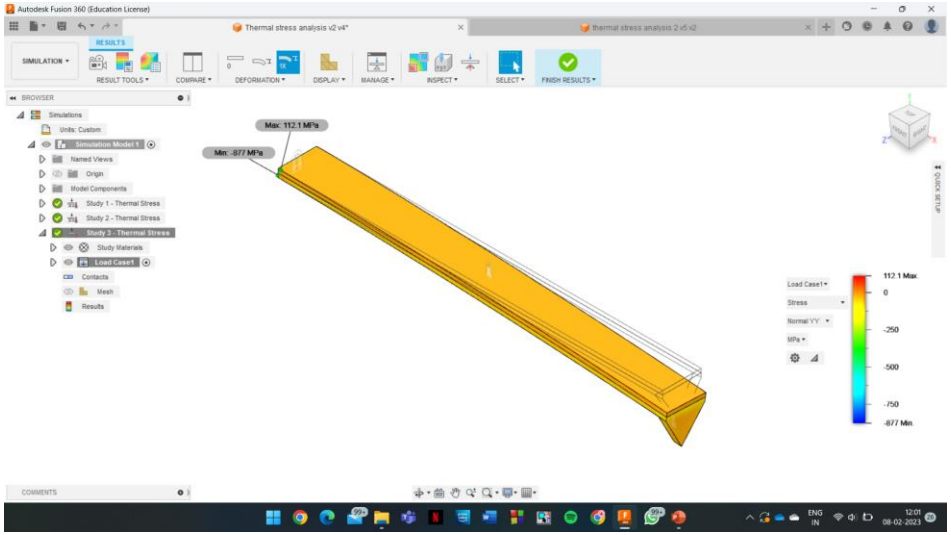


Load case 2

EX03 - Thermal stress analysis of bimetallic strip

Syy distribution Pattern comparison between Load case 1 & 2 (Indicating Max/Min value)



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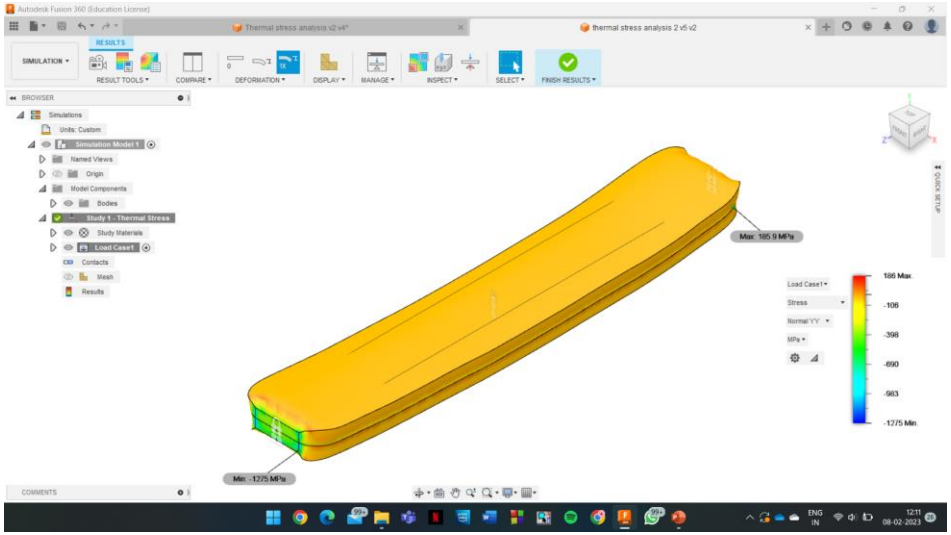
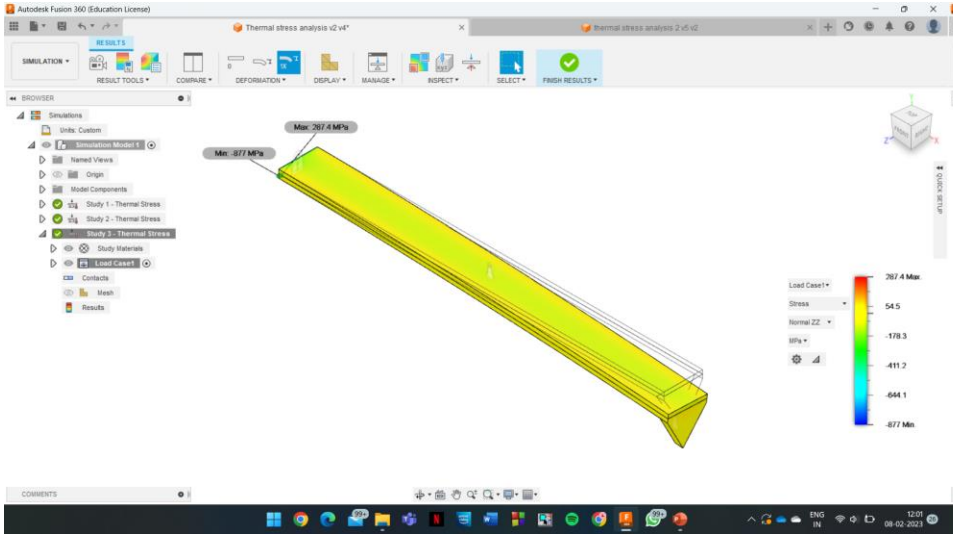


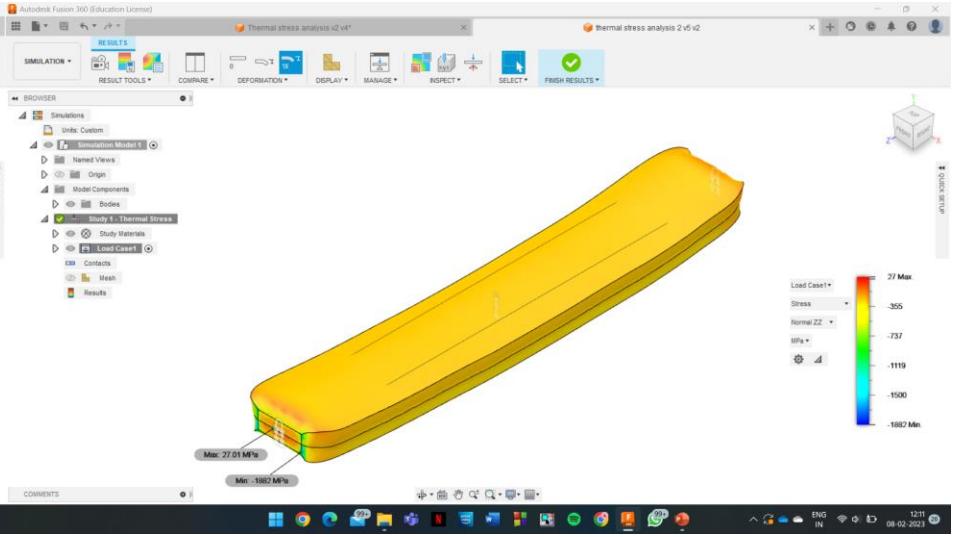
Load case 2

EX03 - Thermal stress analysis of bimetallic strip

Szz distribution Pattern comparison between Load case 1 & 2 (Indicating Max/Min value)



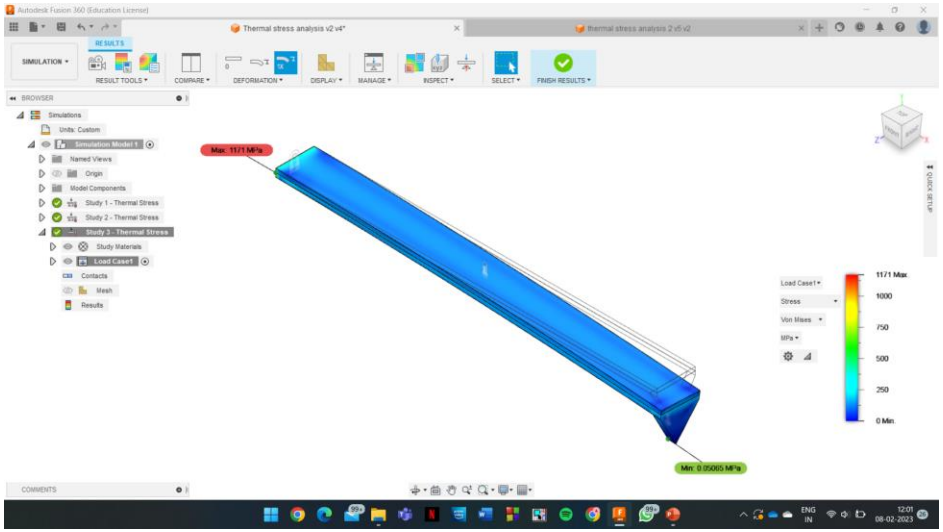
Load case 1



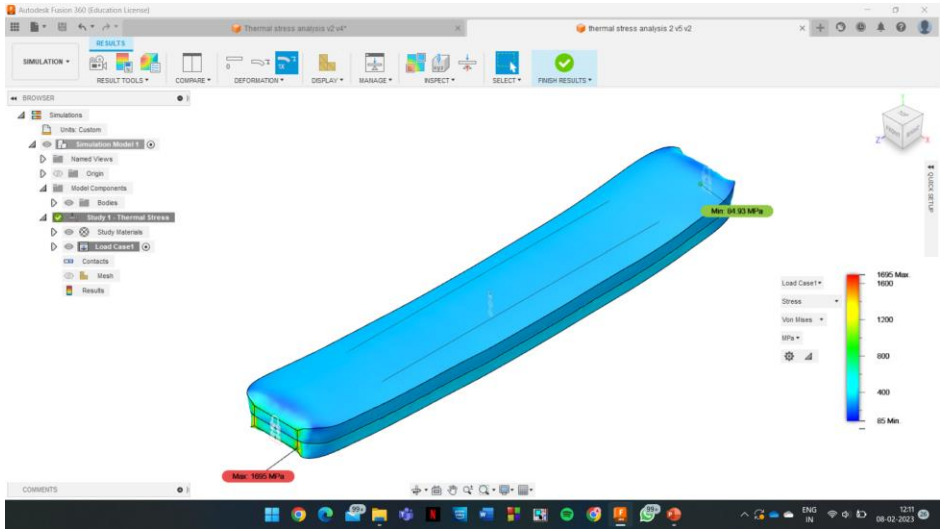
Load case 2

EX03 - Thermal stress analysis of bimetallic strip

Von Mises stress distribution Pattern comparison between Load case 1 & 2 (Indicating Max/Min value)



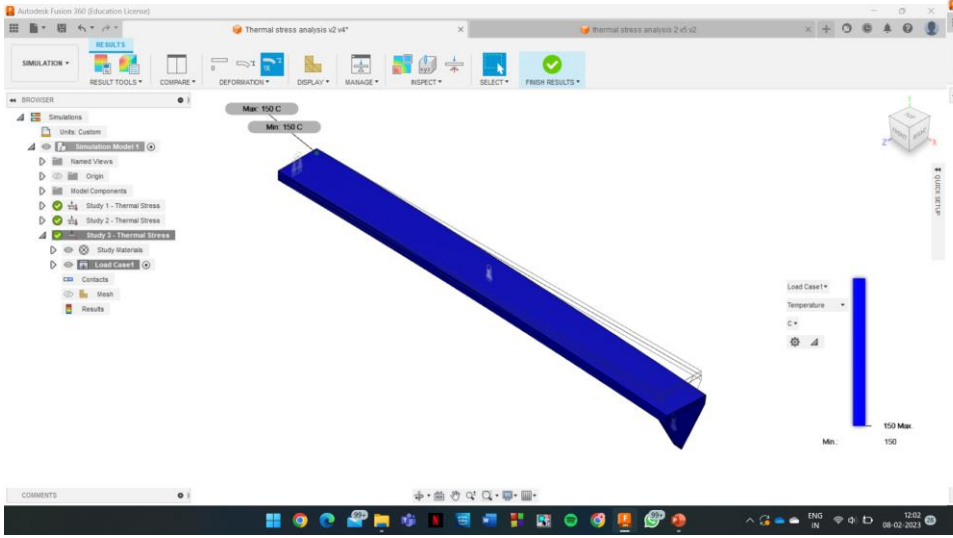
Load case 1



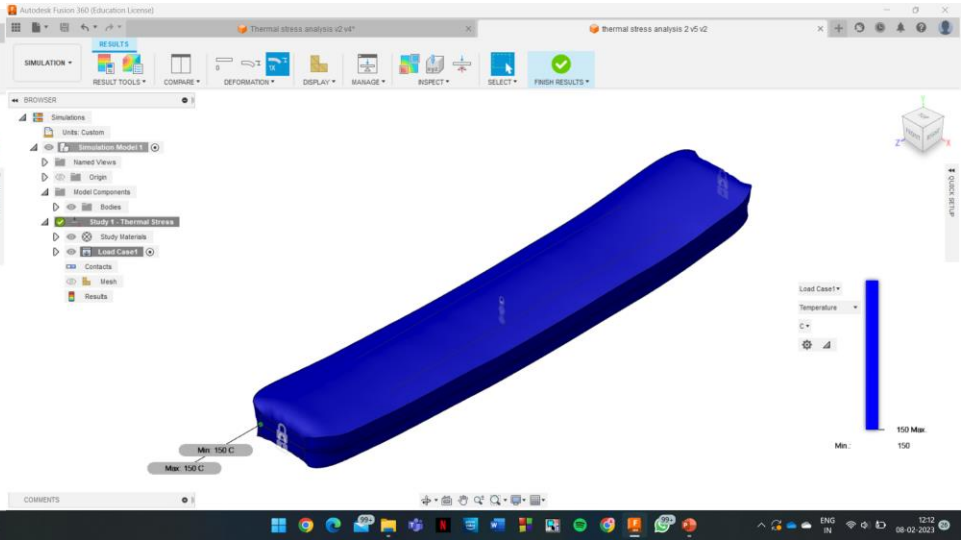
Load case 2

EX03 - Thermal stress analysis of bimetallic strip

Temperature distribution Pattern comparison between Load case 1 & 2 (Indicating Max/Min value)



Load case 1



Load case 2

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Inference and Conclusion

The thermal analysis of the bi-metallic strip is done. This gives the full analysis about the stress that acts on the strip.

20BME0699
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